PREP 2, LESSONS 3–4

Density

$\rho = \frac{m}{V}$; measuring the density of solids and liquids (required practical)

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the function of transformers. DN 148 [1]

To increase or decrease the potential difference.

2 Describe the motion of particles in a solid. DN 170 [1]

They vibrate around a fixed position.

3 Describe the cause of gas pressure. DN 212 [1]

Gas particles colliding with the walls of the container.

Key ideas LESSONS 3–4

Show your working

4 A block of aluminium has a mass of 540 g and a volume of 200 cm³. Use $\rho = \frac{m}{V}$

Calculate the density of aluminium in g/cm³. [3]

$\rho = \frac{m}{V}$ ✓

$\rho = \frac{540}{200}$ ✓

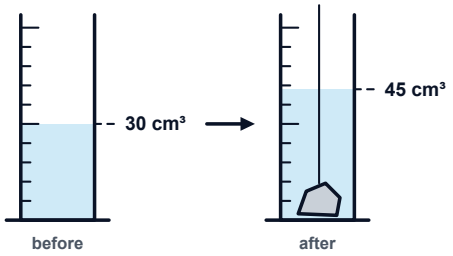
$\rho = 2.7$ g/cm³ ✓

5 A student lowers a stone into a measuring cylinder of water.

a Calculate the volume of the stone. [1]

45 − 30 = 15 cm³ ✓

b The scale is read at the top (bottom) of the meniscus. [1]



6 Explain why a gas has a much lower density than a liquid. [2]

The particles in a gas are much further apart ✓

so there is less mass in the same volume ✓